MEKF With Navigation Frame Attitude Error Parameterization for INS/GPS

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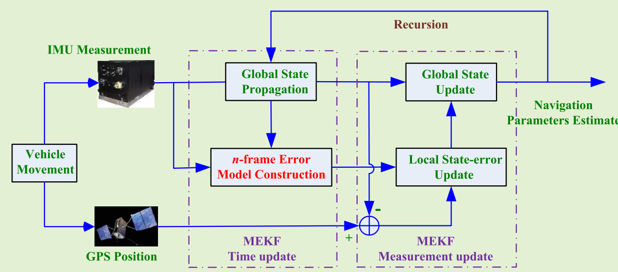
***Abstract*—The multiplicative extended Kalman filter (MEKF) is an attractive method for inertial navigation system and Global Positioning System (INS/GPS) integration due to its flexibility and computational efficiency. Traditionally, in MEKF, the attitude error that is used as the local filtering state is always expressed in the body frame. With such definition, the resulting attitude error model contains inevitable disturbance, which may degrade the filtering performance. Motived by the aforementioned fact, this work revisits the MEKF for INS/GPS with the utilization of navigation frame attitude error model in the filter. The presentation of MEKF in this paper is in a self-contained manner, which is tutorial in nature and clearly defines all involved attitude and attitude errors for frame transformation. Since there is no noise/disturbance contaminated factor in the model, the proposed model possesses advantages in terms of both robustness and accuracy. The subtle differences that arise between the used navigation frame attitude error model and traditional body frame attitude error model in MEKF are discussed and evaluated based on simulation and field test results.**



***Index Terms*—Attitude estimation, extended Kalman filter (EKF), inertial navigation, inertial navigation system and Global Positioning System (INS/GPS).**

I. Introduction

THE multiplicative extended Kalman filter (MEKF) is a computationally efficient adaptation of the extended Kalman filter (EKF) to the problem of attitude state estimation with quaternion as the attitude parameterization. The term "multiplicative" means that the posteriori quaternion is updated using quaternion multiplication, which can preserve the quaternion unit norm constraint to machine accuracy. MEKF has been the most prolific approach to attitude estimation applications, such as spacecraft attitude estimation, relative navigation, inertial navigation system and Global Positioning System integration (INS/GPS) [1]–[5]. MEKF has been regarded as the workhorse of on-board spacecraft attitude estimation. Actually, the MEKF is just pioneered and popularized in field of spacecraft attitude estimation [1], [6]–[10]. In the fundamental work of MEKF [1], the multiplicative structure and dimensional reduction for attitude covariance have been well studied. This work, to one extent or another, promotes researchers and engineers recognizing the utility of employing quaternions. Shuster has surveyed the attitude representations which can be used as local attitude errors for MEKF, such as the rotation angle vector, two times the vector of Rodrigues parameters, two times the vector part of the quaternion, and so on. Shuster prefers to use two times the vector part of the quaternion as local attitude error for MEKF, while Markley prefers to use Rodrigues parameters. In [6], Markley discussed the conceptual advantage of the Rodrigues parameters for MEKF. In [2], the second-order version of MEKF is given by Markley, which is similar to the extension of EKF to second-order EKF. In [11], a geometric EKF is developed for attitude estimation employing common frame error representations. The geometric attitude estimator in [11] is derived using the covariance projection approach, which is different with the MEKF. However, the re-derivation of this geometric attitude estimator using MEKF approach has been presented in [12]. In [3], the MEKF is used for relative spacecraft attitude and position estimation. The quaternion is used to describe the relative kinematics and the two times the vector part of the quaternion is used as the local attitude error for filtering, which is similar with that in [1]. In [13], the covariance modification for the reset operation in MEKF is studied. However, Markley asserted that ignoring such modification will not produce any problems in applications [6], [14] gives a tutorial for quaternion algebra, which is the foundation for derivation of MEKF. Some excellent monographs have also discussed the derivation of MEKF, as shown in [15], [16]. In [9], the MEKF is extended

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to unscented quaternion estimator (USQUE) for spacecraft attitude estimation. USQUE has learned many lessons from MEKF in terms of how to preserve the quaternion unit norm constraint. USQUE can be regarded as the multiplicative form of the traditional unscented Kalman filter (UKF) for attitude estimation. The multiplicative structure of MEKF and USQUE means that the quaternion is used as the global attitude parameterization, while a three-dimensional attitude representation is used to parameterize the local attitude error. The USQUE can be viewed as the most competitive and representative nonlinear attitude estimator. The USQUE is designed detailedly for INS/GPS application in [5]. In [5], the MEKF for INS/GPS application has also been presented as comparison. Actually, the structure of MEKF and USQUE has also inspired many other sophisticated filters for attitude estimation applications [7], [8], [10]. However, the MEKF is still widely regarded as the workhorse due to its flexibility and computational efficiency. Meanwhile, most of the extensions of these sophisticated filters for attitude estimation are straightforward and much less works are devoted to the investigation of filtering structure and filtering model.

This work will not trace the extension of the MEKF structure to more sophisticated filters, but instead this work is devoted to revisiting the filtering model for MEKF. Through careful analysis, it can be found that in the aforementioned applications of MEKF, the local attitude errors are all defined in the body frame. In [17], the attitude error is defined in navigation frame for MEKF which is applied to star sensor based satellite attitude and attitude rate estimate. However, the difference with traditional body frame attitude error model has not been discussed in [17]. In field of robot, the MEKF with reference frame attitude error has been studied in recent few years. It is argued that the filtering consistency can be improved with reference frame attitude error model when compared with body frame attitude error model [18]–[21]. In [4], the MEKF with reference frame attitude error is applied to estimate the relative state of a multi-rotor vehicle. In [22], both the attitude error models in body and reference frames are derived for error-state Kalman filters, which is suited for real applications using integration of signals from an inertial measurement unit. It should be noted that in these reference frame attitude error models, the coupling between attitude and velocity has not been considered. Although [4] argues that their MEKF contains the full state (position, velocity, attitude, accelerometer biases, and gyroscope biases) of an unmanned aircraft system, the transport rate in attitude kinematic caused by velocity has also been ignored. In [23], [24], the geometric EKF proposed in [11] is extended for INS/GPS application. In this extension, the attitude errors in both body and reference frames are considered. However, when deriving the detailed attitude error model, the reference frame attitude error is reconstructed through direction cosine matrix (DCM) transformation of the body frame attitude error. Actually, in [24] the attitude error model in reference frame (navigation frame specially for INS/GPS) is derived through direct attitude transformation of the attitude error model in body frame. However, according to the derivation in this work, it is shown that there is no such direct attitude transformation relationship between the attitude error models in body and navigation frame.

It can be concluded from the aforementioned review that there are two types of attitude error definitions in MEKF, that is, reference frame attitude error and body frame attitude error. In the traditional MEKF for spacecraft attitude estimation or INS/GPS, local attitude error is expressed in body frame. It is shown in [5] that there is noise/disturbance contaminated factor (body frame angular rate provided by gyroscopes) in the body frame attitude error model, which will therefore degrade the corresponding filtering performance. Moreover, to the best knowledge of the authors there has been no investigation on MEKF with reference frame attitude error for INS/GPS. With this consideration, we would therefore like to derive the reference frame attitude error model for MEKF application in INS/GPS. The underlying expectation is to improve INS/GPS integration performance through model modification. Some of aforementioned valuable works are of similar scope to the current work, but this paper still provides several meaningful extensions. First, the navigation frame attitude error model is derived in detail, which is the foundation for the corresponding MEKF application. Both the transport rate caused by linear velocity and earth rotation rate are considered when deriving the attitude error model, which is the main difference with that in [4]. Meanwhile, the navigation frame attitude error model is derived from the starting attitude error definition. This is the main difference with that in [1], [2], [5]. Moreover, it is revealed that there is no such direct attitude transformation relationship between the attitude error models in body and navigation frame as argued in [24]. Second, all the involved attitude errors (when parameterized by DCM or quaternion) satisfy the multiplication chain rule. During the attitude error model derivation, the attitude transformation from one frame to another is marked explicitly for both attitude and attitude error, which can facilitate the understanding of attitude error definition and attitude error model derivation. With such scenario, the navigation frame or body frame attitude error model can be defined explicitly consistent with their intrinsic meaning. Third, the performance of the MEKF with body and navigation frame attitude errors for INS/GPS is evaluated comprehensively using simulated and field test data. The test results indicate that the proposed navigation frame attitude error model is more preferred due to its accuracy and robustness.

The remainder of the paper is organized as follows: Section II briefly introduces the necessary coordinate frames and nomenclature. Section III is devoted to designing the MEKF with navigation frame attitude error model for INS/GPS application. In Section IV, performance evaluation of the proposed algorithm is carried out using simulated and field test data with main comparison with MEKF with body frame attitude error model. Main conclusions are drawn in Section V.

II. Coordinate Frames and Nomenclature

The coordinate frames involved in this paper mainly include local level navigation frame denoted as n, INS body frame denoted as b, Earth-centered Earth-fixed frame denoted as e, the fixed inertial frame denoted as i and the calculated navigation frame denoted as n'.

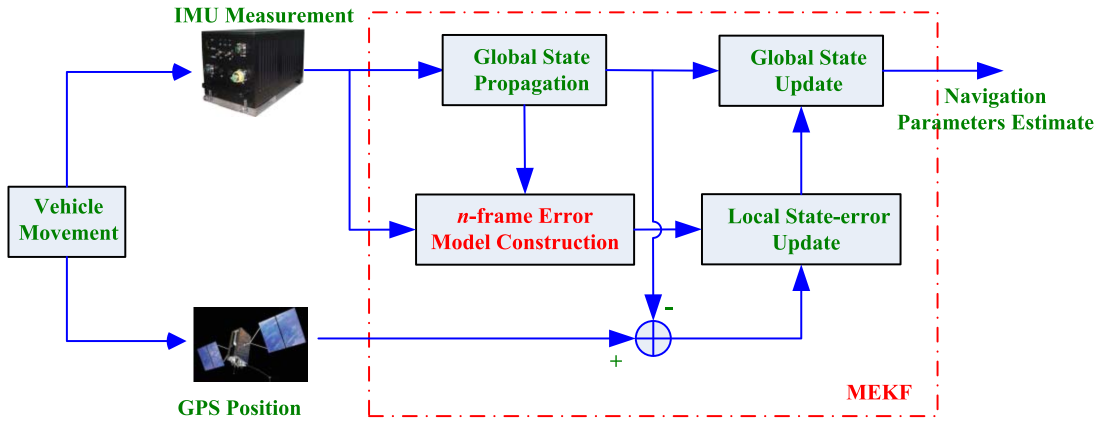

Fig. 1. The structure of the investigated method.

Throughout this paper, the vector $\tilde{\mathbf{u}}$ denotes the error-contaminated value corresponding to the error-free vector $\mathbf{u}$. The error may be caused by some arithmetic error or by the measured noise and bias. $\hat{\mathbf{u}}$ denotes the estimate value using some filtering methods. $\mathbf{x}_{ba}^{c}$ denotes the angular rate vector of frame a with respect to frame b, expressed in frame c. $\mathbf{d}^{c}$ denotes the vector expressed in frame c. $(\mathbf{u}\times)$ denotes the skew symmetric matrix corresponding to vector $\mathbf{u}$. $\otimes$ denotes the quaternion multiplication. $\mathbf{q}^{*}$ denotes the conjugate quaternion corresponding to $\mathbf{q}$. $\mathbf{q}_{a}^{b}$ denotes the attitude quaternion from frame a to frame b.

During the attitude error model derivation, all the involved attitude or attitude error parameterizations satisfy the multiplication chain rule. For example, $\mathbf{q}_{a}^{b} = \mathbf{q}_{c}^{b} \otimes \mathbf{q}_{a}^{c}$.

III. Formulation of MEKF With Navigation Frame Attitude Error Model

The structure of the investigated method is illustrated in Fig. 1. In the following subsections, the involved block diagram will be explained explicitly.

A. System Model of the Inertial-Based Integration Navigation

For inertial-based integration navigation, the dynamic model can be formulated by the INS standard equations. Denote the attitude quaternion from b frame to n frame as $\mathbf{q}_{b}^{n}$, velocity as $\mathbf{v}^{n} = \left[\mathbf{v}_{E}^{n} \quad \mathbf{v}_{N}^{n} \quad \mathbf{v}_{U}^{n}\right]^{T}$ and position as $\mathbf{p} = [L \ \lambda \ h]^{T}$ with L being latitude, λ longitude and h height. If the constant drift biases of the inertial sensors are included in the state, the corresponding model is given by [25]

$$\begin{cases} \dot{\mathbf{q}}_{b}^{n} = \dfrac{1}{2}\mathbf{q}_{b}^{n} \otimes \left[\tilde{\boldsymbol{\omega}}_{ib}^{b} - \boldsymbol{\varepsilon}^{b} - \mathbf{C}\left(\mathbf{q}_{b}^{n}\right)^{T} \boldsymbol{\omega}_{in}^{n}\right] \\ \dot{\mathbf{v}}^{n} = \mathbf{C}\left(\mathbf{q}_{b}^{n}\right)\left(\tilde{\mathbf{f}}^{b} - \nabla^{b}\right) - \left(2\boldsymbol{\omega}_{ie}^{n} + \boldsymbol{\omega}_{en}^{n}\right) \times \mathbf{v}^{n} + \mathbf{g}^{n} \\ \dot{\mathbf{p}}^{n} = \mathbf{R}_{c}\mathbf{v}^{n} \\ \dot{\boldsymbol{\varepsilon}}^{b} = \mathbf{0}_{3\times 1} \\ \dot{\nabla}^{b} = \mathbf{0}_{3\times 1} \end{cases} \tag{1}$$

where

$$\boldsymbol{\omega}_{in}^{n} = \boldsymbol{\omega}_{ie}^{n} + \boldsymbol{\omega}_{en}^{n} \tag{2}$$

$$\boldsymbol{\omega}_{ie}^{n} = \left[0 \quad \omega_{ie}\cos L \quad \omega_{ie}\sin L\right]^{T} \tag{3}$$

$$\boldsymbol{\omega}_{en}^{n} = \left[-\frac{\mathbf{v}_{N}^{n}}{R_{M}+h} \quad \frac{\mathbf{v}_{E}^{n}}{R_{N}+h} \quad \frac{\mathbf{v}_{E}^{n}}{R_{N}+h}\tan L\right]^{\mathrm{T}} \tag{4}$$

$$\mathbf{R}_{c} = \begin{bmatrix} 0 & -1/(R_{M}+h) & 0 \\ \sec L/(R_{N}+h) & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5}$$

R_{N} and R_{M} are the transverse radius of curvature and the meridian radius of curvature of the WGS-84 reference ellipsoid, respectively. $\tilde{\boldsymbol{\omega}}_{ib}^{b}$ is the error-contaminated body angular rate provided by gyroscopes and $\boldsymbol{\varepsilon}^{b}$ is the corresponding constant drift bias. $\tilde{\mathbf{f}}^{b}$ is the outputs of the accelerometers and ∇^{b} is the corresponding constant drift bias. $\mathbf{C}\left(\mathbf{q}_{b}^{n}\right)$ is the DCM corresponding to $\mathbf{q}_{b}^{n}$. Denote the quaternion as

$$\mathbf{q}_{b}^{n} = \left[\mathbf{q}_{0} \quad \mathbf{q}_{\mathbf{v}}^{T}\right]^{T} = \left[\mathbf{q}_{0} \quad \mathbf{q}_{1} \quad \mathbf{q}_{2} \quad \mathbf{q}_{3}\right]^{T} \tag{6}$$

where $\mathbf{q}_{0}$ is the scalar part and $\mathbf{q}_{\mathbf{v}}$ is the vector part. The corresponding DCM is given by

$$\mathbf{C}\left(\mathbf{q}_{b}^{n}\right) = \mathbf{q}_{\mathbf{v}}\mathbf{q}_{\mathbf{v}}^{T} + \mathbf{q}_{0}^{2}\mathbf{I}_{3\times 3} + 2\mathbf{q}_{0}\left[\mathbf{q}_{\mathbf{v}}\times\right] + \left[\mathbf{q}_{\mathbf{v}}\times\right]^{2} \tag{7}$$

B. State Transition Matrix Derivation

If we treat the inertial-based integration navigation as a general dynamic system, the state vector corresponding to (1) is given by

$$\mathbf{x} = \left[\mathbf{q}_{b}^{nT} \quad \mathbf{v}^{nT} \quad \mathbf{p}^{nT} \quad \boldsymbol{\varepsilon}^{bT} \quad \nabla^{bT}\right]^{T} \tag{8}$$

It is clearly shown that the state model (1) is nonlinear. In order to apply the EKF, the state transition matrix should be derived, which can be accomplished through taking partial derivative as

$$\mathbf{A}=\begin{bmatrix} \frac{\partial \dot{\mathbf{q}}_b^n}{\partial \mathbf{q}_b^n} & \frac{\partial \dot{\mathbf{q}}_b^n}{\partial \mathbf{v}^n} & \frac{\partial \dot{\mathbf{q}}_b^n}{\partial \mathbf{p}^n} & \frac{\partial \dot{\mathbf{q}}_b^n}{\partial \boldsymbol{\varepsilon}^b} & \mathbf{0}_{4\times 3} \\ \frac{\partial \dot{\mathbf{v}}^n}{\partial \mathbf{q}_b^n} & \frac{\partial \dot{\mathbf{v}}^n}{\partial \mathbf{v}^n} & \frac{\partial \dot{\mathbf{v}}^n}{\partial \mathbf{p}^n} & \mathbf{0}_{3\times 3} & \frac{\partial \dot{\mathbf{v}}^n}{\partial \nabla^b} \\ \mathbf{0}_{3\times 4} & \frac{\partial \dot{\mathbf{p}}^n}{\partial \mathbf{v}^n} & \frac{\partial \dot{\mathbf{p}}^n}{\partial \mathbf{p}^n} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 4} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 4} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \end{bmatrix} \tag{9}$$

Although the EKF can be readily applied with system model (1) and state transition matrix (9), there are some drawbacks in such application. On the one hand, there is redundancy for quaternion formulation of three-dimensional attitude, which makes the covariance corresponding to quaternion be not full rank. This can cause significant difficulties during the filtering process. On the other hand, the 4×4 covariance propagation and update for quaternion will consume more computational burden than the following designed MEKF.

With the aforementioned consideration, the MEKF is designed in a more sophisticated manner. In MEKF there are two types of state vector, that is, one is global state vector for state propagation and update and the other is local state-error vector for covariance propagation and update. The global state and local state-error vector are defined as

$$\mathbf{x}=\begin{bmatrix} \mathbf{q}_{bn}^T & \mathbf{v}^{nT} & \mathbf{p}^{nT} & \boldsymbol{\varepsilon}^{bT} & \nabla^{bT} \end{bmatrix}^T \tag{10}$$

$$\boldsymbol{\delta}\mathbf{x}=\begin{bmatrix} \boldsymbol{\alpha}^T & \boldsymbol{\delta}\mathbf{v}^{nT} & \boldsymbol{\delta}\mathbf{p}^{nT} & \boldsymbol{\delta\varepsilon}^{bT} & \boldsymbol{\delta}\nabla^{bT} \end{bmatrix}^T \tag{11}$$

where $\boldsymbol{\alpha}$ is the attitude error in Euler angle form. The state-error vector used in MEKF is defined as "true value *subtract* estimate value", which is consistent with that in EKF. The error-dynamics used in the MEKF propagation can be given by

$$\boldsymbol{\delta}\dot{\mathbf{x}}=\bar{\mathbf{A}}\boldsymbol{\delta}\mathbf{x}+\mathbf{G}\mathbf{w} \tag{12}$$

where $\mathbf{w}$ is the process noise and $\mathbf{G}$ is the corresponding transition matrix which will be presented in the following context. $\bar{\mathbf{A}}$ is the state transition matrix for local state-error vector, which can be derived as

$$\bar{\mathbf{A}}=\begin{bmatrix} \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \boldsymbol{\alpha}} & \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \boldsymbol{\delta}\mathbf{v}^n} & \frac{\partial \dot{\boldsymbol{\alpha}}^{\,n}}{\partial \boldsymbol{\delta}\mathbf{p}} & \frac{\partial \dot{\boldsymbol{\alpha}}}{\partial \boldsymbol{\delta\varepsilon}^b} & \mathbf{0}_{3\times 3} \\ \frac{\partial \boldsymbol{\delta}\dot{\mathbf{v}}^n}{\partial \boldsymbol{\alpha}} & \frac{\partial \boldsymbol{\delta}\dot{\mathbf{v}}^n}{\partial \boldsymbol{\delta}\mathbf{v}^n} & \frac{\partial \boldsymbol{\delta}\dot{\mathbf{v}}^{n\,n}}{\partial \boldsymbol{\delta}\mathbf{p}} & \mathbf{0}_{3\times 3} & \frac{\partial \boldsymbol{\delta}\dot{\mathbf{v}}^n}{\partial \boldsymbol{\delta}\nabla^b} \\ \mathbf{0}_{3\times 3} & \frac{\partial \boldsymbol{\delta}\dot{\mathbf{p}}^n}{\partial \boldsymbol{\delta}\mathbf{v}^n} & \frac{\partial \boldsymbol{\delta}\dot{\mathbf{p}}^n}{\partial \boldsymbol{\delta}\mathbf{p}^n} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} & \mathbf{0}_{3\times 3} \end{bmatrix} \tag{13}$$

In order to derive the state transition matrix $\bar{\mathbf{A}}$, we should firstly derive the differential equations of the local state-error vector.

The quaternion error is defined in the multiplicative form as

$$\boldsymbol{\delta}\mathbf{q}=\mathbf{q}_b^n \otimes \hat{\mathbf{q}}_b^{n^*} \tag{14}$$

where $\hat{\mathbf{q}}_b^n$ is the quaternion estimate by MEKF. From (14) we can see why the attitude error is regarded as error expressed in navigation frame. $\hat{\mathbf{q}}_b^n$ can be rewritten as $\mathbf{q}_b^{n'}$, where n' can be viewed as the navigation frame with calculated errors. According to the chain rule of the quaternion, $\boldsymbol{\delta}\mathbf{q}$ can be expanded as $\boldsymbol{\delta}\mathbf{q}=\mathbf{q}_b^n \otimes \mathbf{q}_{n'}^b=\mathbf{q}_{n'}^n$ which is just the attitude difference between the navigation frame and calculated navigation frame.

The derivative of $\boldsymbol{\delta}\mathbf{q}$ can be derived as

$$\boldsymbol{\delta}\dot{\mathbf{q}}=\dot{\mathbf{q}}_b^n \otimes \hat{\mathbf{q}}_b^{n*}+\mathbf{q}_b^n \otimes \dot{\hat{\mathbf{q}}}_b^{n^*} \tag{15}$$

The differential equation of $\hat{\mathbf{q}}_b^n$ is given by

$$\dot{\hat{\mathbf{q}}}_b^n=\frac{1}{2}\hat{\mathbf{q}}_b^n \otimes \hat{\boldsymbol{\omega}}_{nb}^b \tag{16}$$

where

$$\hat{\boldsymbol{\omega}}_{nb}^b=\hat{\boldsymbol{\omega}}_{ib}^b-\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\hat{\boldsymbol{\omega}}_{in}^n \tag{17}$$

Substituting (16) into (15) gives

$$\begin{aligned}\boldsymbol{\delta}\dot{\mathbf{q}}&=\frac{1}{2}\mathbf{q}_b^n \otimes \boldsymbol{\omega}_{nb}^b \otimes \hat{\mathbf{q}}_b^{n^*}+\frac{1}{2}\mathbf{q}_b^n \otimes \left(\hat{\mathbf{q}}_b^n \otimes \hat{\boldsymbol{\omega}}_{nb}^b\right)^*\\ &=\frac{1}{2}\mathbf{q}_b^n \otimes \hat{\mathbf{q}}_b^{n^*} \otimes \hat{\mathbf{q}}_b^n \otimes \left(\boldsymbol{\omega}_{nb}^b-\hat{\boldsymbol{\omega}}_{nb}^b\right)\hat{\mathbf{q}}_b^{n^*}\\ &=\frac{1}{2}\boldsymbol{\delta}\mathbf{q} \otimes \left[\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\boldsymbol{\delta\omega}_{nb}^b\right]\end{aligned} \tag{18}$$

$\boldsymbol{\delta\omega}_{nb}^b$ is clearly defined in (18) as

$$\boldsymbol{\delta\omega}_{nb}^b=\boldsymbol{\omega}_{nb}^b-\hat{\boldsymbol{\omega}}_{nb}^b \tag{19}$$

In order to derive the detailed form of $\boldsymbol{\delta\omega}_{nb}^b$, we firstly define the following error vectors:

$$\boldsymbol{\delta\omega}_{ib}^b=\boldsymbol{\omega}_{ib}^b-\hat{\boldsymbol{\omega}}_{ib}^b \tag{20}$$

$$\boldsymbol{\delta\omega}_{in}^n=\boldsymbol{\omega}_{in}^n-\hat{\boldsymbol{\omega}}_{in}^n \tag{21}$$

Substituting (17) and (20), (21) into (19) gives

$$\begin{aligned}\boldsymbol{\delta\omega}_{nb}^b&=\boldsymbol{\omega}_{nb}^b-\hat{\boldsymbol{\omega}}_{nb}^b\\ &=\boldsymbol{\omega}_{ib}^b-\left[\mathbf{C}\left(\boldsymbol{\delta}\mathbf{q}\right)\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\right]^T\boldsymbol{\omega}_{in}^n-\left[\hat{\boldsymbol{\omega}}_{ib}^b-\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\hat{\boldsymbol{\omega}}_{in}^n\right]\\ &\approx \boldsymbol{\delta\omega}_{ib}^b-\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\left(\mathbf{I}_{3\times 3}-\boldsymbol{\alpha}\times\right)\left(\hat{\boldsymbol{\omega}}_{in}^n+\boldsymbol{\delta\omega}_{in}^n\right)\\ &\quad+\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\hat{\boldsymbol{\omega}}_{in}^n\\ &\approx -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\hat{\boldsymbol{\omega}}_{in}^n\times\boldsymbol{\alpha}-\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T\boldsymbol{\delta\omega}_{in}^n+\boldsymbol{\delta\omega}_{ib}^b\end{aligned} \tag{22}$$

From the third line to fourth line of (22), we have used the following approximation

$$\mathbf{C}\left(\boldsymbol{\delta}\mathbf{q}\right)\approx \mathbf{I}_{3\times 3}+\left(\boldsymbol{\alpha}\times\right) \tag{23}$$

which is only hold for small Euler angle errors.

Denote $\boldsymbol{\delta\omega}_{nb}^n=\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\boldsymbol{\delta\omega}_{nb}^b$ and its detailed form can be given by

$$\boldsymbol{\delta\omega}_{nb}^n=\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\boldsymbol{\delta\omega}_{nb}^b=-\hat{\boldsymbol{\omega}}_{in}^n\times\boldsymbol{\alpha}-\boldsymbol{\delta\omega}_{in}^n+\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\boldsymbol{\delta\omega}_{ib}^b \tag{24}$$

Substituting (24) into (18) gives

$$\boldsymbol{\delta}\dot{\mathbf{q}}=\frac{1}{2}\begin{bmatrix} 0 & -\boldsymbol{\delta\omega}_{nb}^{nT} \\ \boldsymbol{\delta\omega}_{nb}^n & -\left(\boldsymbol{\delta\omega}_{nb}^n\times\right) \end{bmatrix}\boldsymbol{\delta}\mathbf{q} \tag{25}$$

With small attitude error assumption, $\boldsymbol{\delta}\mathbf{q}$ can be approximated as

$$\boldsymbol{\delta}\mathbf{q}=\begin{bmatrix} 1 & \boldsymbol{\alpha}^T/2 \end{bmatrix}^T \tag{26}$$

According to (25) and (26) we can derive the differential equation of $\boldsymbol{\alpha}$ as

$$\dot{\boldsymbol{\alpha}} = \boldsymbol{\delta\omega}_{nb}^{n} = -\hat{\boldsymbol{\omega}}_{in}^{n} \times \boldsymbol{\alpha} - \boldsymbol{\delta\omega}_{in}^{n} + \mathbf{C}\left(\hat{\mathbf{q}}_{b}^{n}\right)\boldsymbol{\delta\omega}_{ib}^{b} \tag{27}$$

In (27), $\boldsymbol{\delta\omega}_{in}^{n} = \boldsymbol{\delta\omega}_{ie}^{n} + \boldsymbol{\delta\omega}_{en}^{n}$ and the detailed form of $\boldsymbol{\delta\omega}_{ie}^{n}$ is given by

$$\boldsymbol{\delta\omega}_{ie}^{n} = \begin{bmatrix} 0 \\ -\omega_{ie}\sin L \cdot \delta L \\ \omega_{ie}\cos L \cdot \delta L \end{bmatrix} = \mathbf{M}_1\boldsymbol{\delta}\mathbf{p} \tag{28}$$

where $\boldsymbol{\delta}\mathbf{p} = \begin{bmatrix} \delta L & \delta\lambda & \delta h \end{bmatrix}^T$ and $\mathbf{M}_1$ is given by

$$\mathbf{M}_1 = \begin{bmatrix} 0 & 0 & 0 \\ -\omega_{ie}\sin L & 0 & 0 \\ \omega_{ie}\cos L & 0 & 0 \end{bmatrix} \tag{29}$$

Similarly, $\boldsymbol{\delta\omega}_{en}^{n}$ can be derived through taking partial derivative of $\boldsymbol{\omega}_{en}^{n}$ as

$$\begin{aligned} \boldsymbol{\delta\omega}_{en}^{n} &= \begin{bmatrix} \dfrac{-\boldsymbol{\delta}\mathbf{v}_N^n}{R_M+h} + \dfrac{\mathbf{v}_N^n\delta h}{(R_M+h)^2} \\ \dfrac{\boldsymbol{\delta}\mathbf{v}_E^n}{R_N+h} - \dfrac{\mathbf{v}_E^n\delta h}{(R_N+h)^2} \\ \dfrac{\tan L\cdot\boldsymbol{\delta}\mathbf{v}_E^n}{R_N+h} + \dfrac{\mathbf{v}_E^n\sec^2 L\cdot\delta L}{R_N+h} - \dfrac{\mathbf{v}_E^n\tan L\cdot\delta h}{(R_N+h)^2} \end{bmatrix} \\ &= \mathbf{M}_2\boldsymbol{\delta}\mathbf{v}^n + \mathbf{M}_3\boldsymbol{\delta}\mathbf{p} \end{aligned} \tag{30}$$

where

$$\mathbf{M}_2 = \begin{bmatrix} 0 & -1/(R_M+h) & 0 \\ 1/(R_N+h) & 0 & 0 \\ \tan L/(R_N+h) & 0 & 0 \end{bmatrix} \tag{31}$$

$$\mathbf{M}_3 = \begin{bmatrix} 0 & 0 & \dfrac{\mathbf{v}_N^n}{(R_M+h)^2} \\ 0 & 0 & -\dfrac{\mathbf{v}_E^n}{(R_N+h)^2} \\ \dfrac{\mathbf{v}_E^n\sec^2 L}{R_N+h} & 0 & -\dfrac{\mathbf{v}_E^n\tan L}{(R_N+h)^2} \end{bmatrix} \tag{32}$$

In (27), $\boldsymbol{\delta\omega}_{ib}^{b}$ can be calculated as

$$\begin{aligned} \boldsymbol{\delta\omega}_{ib}^{b} &= \boldsymbol{\omega}_{ib}^{b} - \hat{\boldsymbol{\omega}}_{ib}^{b} \\ &= \left(\tilde{\boldsymbol{\omega}}_{ib}^{b} - \boldsymbol{\varepsilon}^b - \boldsymbol{\eta}_{\boldsymbol{\varepsilon}}\right) - \left(\tilde{\boldsymbol{\omega}}_{ib}^{b} - \hat{\boldsymbol{\varepsilon}}^b\right) = -\left(\boldsymbol{\delta\varepsilon}^b + \boldsymbol{\eta}_{\boldsymbol{\varepsilon}}\right) \end{aligned} \tag{33}$$

where $\boldsymbol{\eta}_{\boldsymbol{\varepsilon}}$ is the gyroscopes measurement noise.

The differential equation of the velocity error can be derived through taking partial derivative of $\dot{\mathbf{v}}^n$ as

$$\boldsymbol{\delta}\dot{\mathbf{v}}^n = \boldsymbol{\delta}\mathbf{f}^n - \left(2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n\right) \times \boldsymbol{\delta}\mathbf{v}^n - \left(2\boldsymbol{\delta\omega}_{ie}^n + \boldsymbol{\delta\omega}_{en}^n\right) \times \mathbf{v}^n \tag{34}$$

where

$$\begin{aligned} \boldsymbol{\delta}\mathbf{f}^n &= \mathbf{C}_b^n\mathbf{f}^b - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\hat{\mathbf{f}}^b \\ &= \left(\mathbf{I}_{3\times3} + \boldsymbol{\alpha}\times\right)\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\left(\hat{\mathbf{f}}^b + \boldsymbol{\delta}\mathbf{f}^b\right) - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\hat{\mathbf{f}}^b \\ &= -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\hat{\mathbf{f}}^b \times \boldsymbol{\alpha} + \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\boldsymbol{\delta}\mathbf{f}^b \end{aligned} \tag{35}$$

Similarly with (33), $\boldsymbol{\delta}\mathbf{f}^b$ can be calculated as

$$\begin{aligned} \boldsymbol{\delta}\mathbf{f}^b &= \mathbf{f}^b - \hat{\mathbf{f}}^b \\ &= \left(\tilde{\mathbf{f}}^b - \nabla^b - \boldsymbol{\eta}_{\nabla}\right) - \left(\tilde{\mathbf{f}}^b - \hat{\nabla}^b\right) = -\left(\boldsymbol{\delta}\nabla^b + \boldsymbol{\eta}_{\nabla}\right) \end{aligned} \tag{36}$$

where $\boldsymbol{\eta}_{\nabla}$ is the gyroscopes measurement noise.

The differential equation of the position error can be derived through taking partial derivative of $\dot{\mathbf{p}}^n$ as

$$\boldsymbol{\delta}\dot{\mathbf{p}}^n = \mathbf{R}_c\boldsymbol{\delta}\mathbf{v}^n + \boldsymbol{\delta}\mathbf{R}_c\mathbf{v}^n \tag{37}$$

where

$$\begin{aligned} \boldsymbol{\delta}\mathbf{R}_c\mathbf{v}^n &= \mathbf{M}_4\boldsymbol{\delta}\mathbf{p}^n \\ &= \begin{bmatrix} 0 & 0 & \dfrac{-\mathbf{v}_N^n}{(R_M+h)^2} \\ \dfrac{\mathbf{v}_E^n\sec L\tan L}{R_N+h} & 0 & -\dfrac{\mathbf{v}_E^n\sec L}{(R_N+h)^2} \\ 0 & 0 & 0 \end{bmatrix}\boldsymbol{\delta}\mathbf{p}^n \end{aligned} \tag{38}$$

where $\mathbf{M}_4$ is clearly defined in (38).

According to differential equations (27), (34) and (37), the elements of state transition matrix $\bar{\mathbf{A}}$ can be readily derived as

$$\partial\dot{\boldsymbol{\alpha}}/\partial\boldsymbol{\alpha} = -\hat{\boldsymbol{\omega}}_{in}^n \tag{39a}$$
$$\partial\dot{\boldsymbol{\alpha}}/\partial\boldsymbol{\delta}\mathbf{v}^n = -\mathbf{M}_2 \tag{39b}$$
$$\partial\dot{\boldsymbol{\alpha}}/\partial\boldsymbol{\delta}\mathbf{p}^n = -\left(\mathbf{M}_1 + \mathbf{M}_3\right) \tag{39c}$$
$$\partial\dot{\boldsymbol{\alpha}}/\partial\boldsymbol{\varepsilon}^b = -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right) \tag{39d}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{v}}^n/\partial\boldsymbol{\alpha} = -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)\left[\hat{\mathbf{f}}^b\times\right] \tag{39e}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{v}}^n/\partial\boldsymbol{\delta}\mathbf{v}^n = \left[-\left(2\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n\right)\times\right] + \left[\mathbf{v}^n\times\right]\mathbf{M}_2 \tag{39f}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{v}}^n/\partial\boldsymbol{\delta}\mathbf{p}^n = \left[\mathbf{v}^n\times\right]\left(2\mathbf{M}_1 + \mathbf{M}_3\right) \tag{39g}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{v}}^n/\partial\boldsymbol{\delta}\nabla^b = -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right) \tag{39h}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{p}}^n/\partial\boldsymbol{\delta}\mathbf{v}^n = \mathbf{R}_c \tag{39i}$$
$$\partial\boldsymbol{\delta}\dot{\mathbf{p}}^n/\partial\boldsymbol{\delta}\mathbf{p}^n = \mathbf{M}_4 \tag{39j}$$

Denote the process noise as $\mathbf{w} = \begin{bmatrix} \boldsymbol{\eta}_{\boldsymbol{\varepsilon}}^T & \boldsymbol{\eta}_{\nabla}^T \end{bmatrix}^T$, $\boldsymbol{\eta}_{\boldsymbol{\varepsilon}}$ is the gyroscopes measurement noise and $\boldsymbol{\eta}_{\nabla}$ the accelerometers measurement noise. The noise transition matrix $\mathbf{G}$ is given by

$$\mathbf{G} = \begin{bmatrix} -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right) & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right) \\ \mathbf{0}_{9\times3} & \mathbf{0}_{9\times3} \end{bmatrix} \tag{40}$$

Since the model in (12) is in continuous-time form, it should be discretized for application of MEKF on computer. The discrete-time state transformation matrix can be calculated as

$$\mathbf{F} = \exp\left(\bar{\mathbf{A}}\cdot dt\right) \tag{41}$$

where dt is the step size of the discretization.

The discrete-time process noise covariance can be calculated as

$$\begin{bmatrix} \mathbf{C} \\ \mathbf{D} \end{bmatrix} = \exp\left[\begin{pmatrix} \bar{\mathbf{A}} & \mathbf{G}\mathbf{Q}_c\mathbf{G} \\ \mathbf{0} & -\bar{\mathbf{A}}^T \end{pmatrix}\cdot dt\right]\cdot\begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix} \tag{42}$$

and $\mathbf{Q} = \mathbf{C}\mathbf{D}^{-1}$. $\mathbf{Q}_c$ is the noise covariance corresponding to $\mathbf{w}$.

If we use the position provided by GPS as the measurements, the corresponding measurement matrix is given by

$$\mathbf{H} = \begin{bmatrix} \mathbf{0}_{3\times6} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times6} \end{bmatrix} \tag{43}$$

C. Recursive MEKF for INS/GPS

The discrete-time MEKF for the INS/GPS can be summarized as follows.

Algorithm 1 MEKF for INS/GPS

Initialization:
$\hat{\mathbf{x}}_0 = E[\mathbf{x}_0]$
$\mathbf{P}_0 = E\left[(\mathbf{x}_0 - \hat{\mathbf{x}}_0)(\mathbf{x}_0 - \hat{\mathbf{x}}_0)^T\right]$
State transition matrix calculation:
$\mathbf{F}_{k-1} = \exp(\bar{\mathbf{A}} \cdot dt)\,|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$
Global state propagation:
$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1})$
Local state-error covariance propagation:
$\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1}\mathbf{P}_{k-1}\mathbf{F}_{k-1}^T + \mathbf{Q}_{k-1}$
Kalman gain calculation:
$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^T\left(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^T + \mathbf{R}_k\right)^{-1}$
Local state-error update:
$\boldsymbol{\delta}\hat{\mathbf{x}}_k = \mathbf{K}_k\left[\mathbf{y}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}\right]$
Local state-error covariance update:
$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}_k\mathbf{H})^T + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^T$
Global state update:
$\hat{\mathbf{x}}_k(1:4) = \hat{\mathbf{x}}_{k|k-1}(1:4) + \frac{1}{2}\boldsymbol{\Xi}\left(\hat{\mathbf{x}}_{k|k-1}(1:4)\right)\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3)$
$\hat{\mathbf{x}}_k(5:16) = \hat{\mathbf{x}}_{k|k-1}(5:16) + \boldsymbol{\delta}\hat{\mathbf{x}}_k(4:15)$

In Algorithm 1, $f(\cdot)$ denotes the compact nonlinear process model of (1). $\mathbf{y}_k$ is the position vector provided by GPS and $\mathbf{R}_k$ is the corresponding measurement noise covariance. With the quaternion definition in (6), $\boldsymbol{\Xi}(\mathbf{q})$ is defined as

$$\boldsymbol{\Xi}(\mathbf{q}) = \begin{bmatrix} -\mathbf{q}_{\mathbf{v}}^T \\ q_0\mathbf{I}_{3\times3} - (\mathbf{q}_{\mathbf{v}}\times) \end{bmatrix} \tag{44}$$

It is seemingly that the quaternion is not updated in multiplicative form as shown in **Algorithm 1**. Actually, the additive quaternion update form is an approximation of the quaternion multiplication. More specifically,

$$\hat{\mathbf{q}}_{b,k}^n = \boldsymbol{\delta}\hat{\mathbf{q}}_{b,k}^n \otimes \hat{\mathbf{q}}_{b,k|k-1}^n \tag{45}$$

where

$$\boldsymbol{\delta}\hat{\mathbf{q}}_{b,k}^n \approx \begin{bmatrix} 1 \\ \frac{1}{2}\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \end{bmatrix} \tag{46}$$

and $\hat{\mathbf{q}}_{b,k|k-1}^n$ is the first four elements of the predicted global state, that is $\hat{\mathbf{q}}_{b,k|k-1}^n = \hat{\boldsymbol{x}}_{k|k-1}(1:4)$. Substituting (46) into (45) gives (47), shown at the bottom of this page, where $\hat{\mathbf{q}}_{b,k|k-1}^n = \begin{bmatrix} \hat{q}_{b,k|k-1,0}^n & \hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^{nT} \end{bmatrix}^T$. From the first line to the second line of (47), the following quaternion multiplication algebra has been used

$$\mathbf{q} = \mathbf{q}_a \otimes \mathbf{q}_b = \begin{bmatrix} q_{b,0} & -\mathbf{q}_{b,\mathbf{v}}^T \\ \mathbf{q}_{b,\mathbf{v}} & q_{b,0}\mathbf{I}_{3\times3} - (\mathbf{q}_{b,\mathbf{v}}\times) \end{bmatrix}\mathbf{q}_a \tag{48}$$

where $\mathbf{q}_b = \begin{bmatrix} q_{b,0} & \mathbf{q}_{b,\mathbf{v}}^T \end{bmatrix}^T$. As there is a first order linearization already inherent in (23) and (26), rigorously updating quaternion using (45) isn't likely to add any additional accuracy compared with (47).

D. Discussion

According to the convention in this paper, the body frame attitude error is defined as

$$\boldsymbol{\delta}\mathbf{q} = \hat{\mathbf{q}}_b^{n*} \otimes \mathbf{q}_b^n \tag{49}$$

where $\hat{\mathbf{q}}_b^n$ can be rewritten as $\mathbf{q}_{b'}^n$, where b' can be viewed as the body frame with calculated errors. According to the chain rule of the quaternion, $\boldsymbol{\delta}\mathbf{q}$ can be expanded as $\boldsymbol{\delta}\mathbf{q} = \mathbf{q}_n^{b'} \otimes \mathbf{q}_b^n = \mathbf{q}_b^{b'}$ which is just the attitude difference between the body frame and calculated body frame.

With the definition (49), the attitude error model can also be derived similarly with that in Section III. B (Detailed derivation can be found in Appendix). The resulting body frame attitude error model is given by

$$\dot{\boldsymbol{\varphi}} = -\hat{\boldsymbol{\omega}}_{ib}^b \times \boldsymbol{\varphi} - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T \boldsymbol{\delta}\boldsymbol{\omega}_{in}^n + \boldsymbol{\delta}\boldsymbol{\omega}_{ib}^b \tag{50}$$

where $\boldsymbol{\varphi}$ is the body frame attitude error in Euler angle form. It can be seen from (27) and (50) that there is no direct attitude transformation relationship between attitude error models in body and navigation frame. Such attitude transformation relationship only holds in static conditions, that is

$$\begin{aligned} \boldsymbol{\omega}_{nb}^b &= \boldsymbol{\omega}_{ib}^b - \mathbf{C}\left(\mathbf{q}_b^n\right)^T \boldsymbol{\omega}_{in}^n = \mathbf{0}_{3\times1} \\ &\Rightarrow \boldsymbol{\omega}_{ib}^b = \mathbf{C}\left(\mathbf{q}_b^n\right)^T \boldsymbol{\omega}_{in}^n \end{aligned} \tag{51}$$

Set aside the performance of the two different attitude error models (they will be evaluated in the next section), the attitude update should be consistent with the corresponding

$$\begin{aligned} \hat{\mathbf{q}}_{b,k}^n &= \begin{bmatrix} 1 \\ \frac{1}{2}\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \end{bmatrix} \otimes \hat{\mathbf{q}}_{b,k|k-1}^n \\ &= \begin{bmatrix} \hat{q}_{b,k|k-1,0}^n & -\hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^{nT} \\ \hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^n & \hat{q}_{b,k|k-1,0}^n \cdot \mathbf{I}_{3\times3} - (\hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^n\times) \end{bmatrix} \begin{bmatrix} 1 \\ \frac{1}{2}\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \end{bmatrix} \\ &= \begin{bmatrix} \hat{q}_{b,k|k-1,0}^n - \frac{1}{2}\hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^{nT}\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \\ \hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^n + \frac{1}{2}\left[\hat{q}_{b,k|k-1,0}^n \cdot \mathbf{I}_{3\times3} - (\hat{\mathbf{q}}_{b,k|k-1,\mathbf{v}}^n\times)\right]\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \end{bmatrix} \\ &= \hat{\mathbf{q}}_{b,k|k-1}^n + \frac{1}{2}\boldsymbol{\Xi}\left(\hat{\mathbf{q}}_{b,k|k-1}^n\right)\boldsymbol{\delta}\hat{\mathbf{x}}_k(1:3) \end{aligned} \tag{47}$$

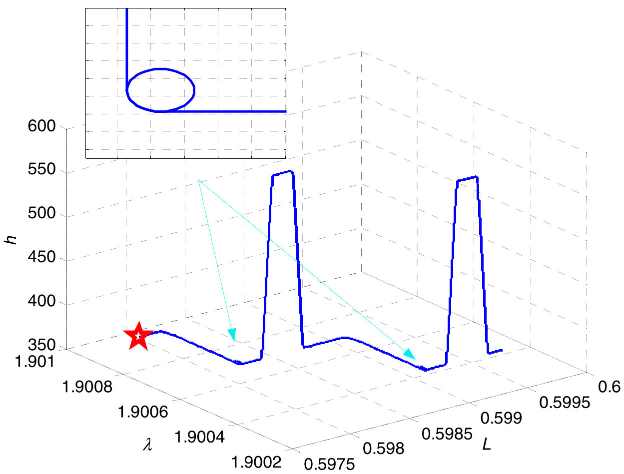

Fig. 2. Trajectory during the simulation.

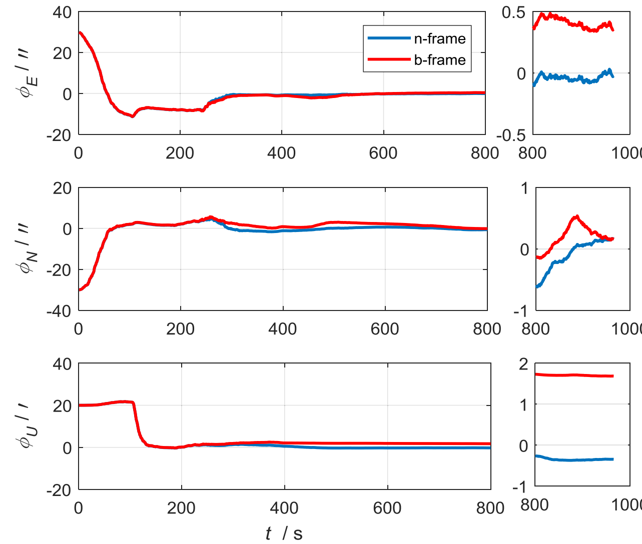

Fig. 3. Attitude estimate errors.

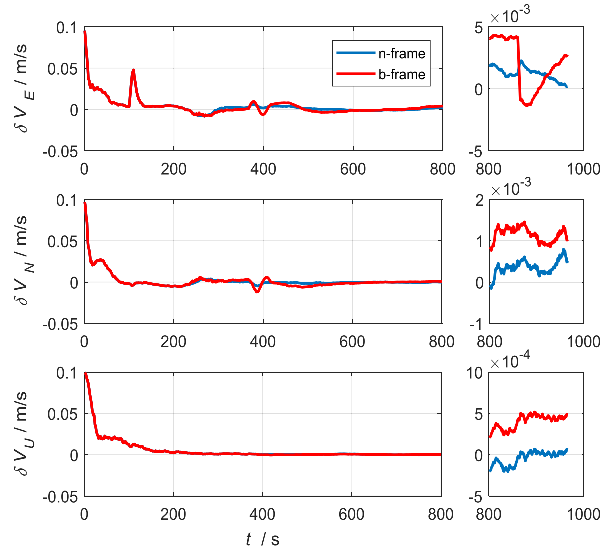

Fig. 4. Velocity estimate errors.

attitude error definition. More specifically, with the navigation frame attitude error definition (14), the attitude update should be performed in the form (45). While for the body frame attitude error definition (49), the corresponding attitude update is performed as

$$\hat{\mathbf{q}}^n_{b,k} = \hat{\mathbf{q}}^n_{b,k|k-1} \otimes \boldsymbol{\delta}\hat{\mathbf{q}}^n_{b,k} \tag{52}$$

In previous works, the expressed frame of the attitude error is always determined according to the position of the attitude error in its definition. For example, in [16], attitude error defined in $\mathbf{q}_{true} = \boldsymbol{\delta}\ \mathbf{q} \otimes \hat{\mathbf{q}}_{estimate}$ is regarded as body frame attitude error, while in $\mathbf{q}_{true} = \hat{\mathbf{q}}_{estimate} \otimes \boldsymbol{\delta}\mathbf{q}$ is regarded as reference frame attitude error. This seems to be just opposite to our definitions in (14) and (49). Such confliction is mainly caused by the different attitude definitions, that is, the attitude quaternion in this paper denotes the attitude from the body frame to reference (navigation) frame while in [5], [16] it denotes the attitude from reference (navigation) frame to body frame. In [17], the reference frame attitude error is defined in a similar manner with that in this paper. In this respect, we cannot arbitrarily determine the expressed frame of the attitude error according to the position of the attitude error in its definition. With this consideration, all the involved attitude and attitude error in this paper are defined with definitely transformation frames.

IV. Simulation and Field Test Results

A. Simulation Results

In order to evaluate the MEKF for INS/GPS with different attitude error definitions, the simulation experiment is carried out. The true trajectory during the simulation is shown in Fig. 2, which contains two circular segments. The five-pointed star denotes the starting point. In this simulation, the constant drift of the gyroscopes is assumed to be $0.03°/h$ and the noise is assumed to be $0.001°/h/\sqrt{Hz}$. The constant drift of the accelerometers is assumed to be $100\mu g$ and the noise is assumed to be$5\mu g/\sqrt{Hz}$. The INS sampling rate is 100 Hz and the GPS sampling rate is 1 Hz. During the integration, the position noise with standard variance $[1, 1, 3]^T\ m$ is added to reflect the single-point GPS measurement quality. For INS/GPS integration, the initial attitude error is set as$[30'', -30'', 20']^T$, the initial velocity error is set as $0.1m/s$ for the three dimensions. The initial position error is set as $[1, 1, 3]^T\ m$.

Firstly, 100 Monte Carlo runs are carried out for the MEKF with different attitude error models. The averaged attitude, velocity and position errors are plotted in Fig. 3-5. The averaged gyroscope and accelerometer drift biases are shown in Fig. 6 and 7, respectively. In Fig. 3-7, the term "b-frame" denotes MEKF with body frame attitude error model and "n-frame" denotes MEKF with navigation frame attitude error model. It is shown in these figures that the MEKF with navigation frame attitude error model performs a litter better than that with body frame attitude error model. Meanwhile, it is shown that the convergent speeds of the two application procedures are almost the same and the performance difference

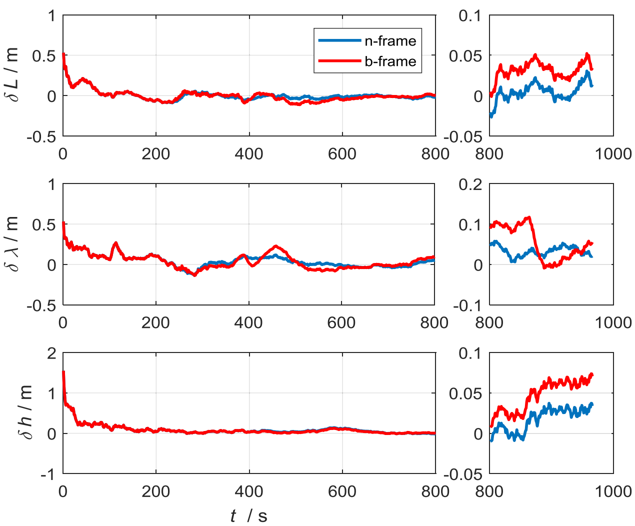

Fig. 5. Position estimate errors.

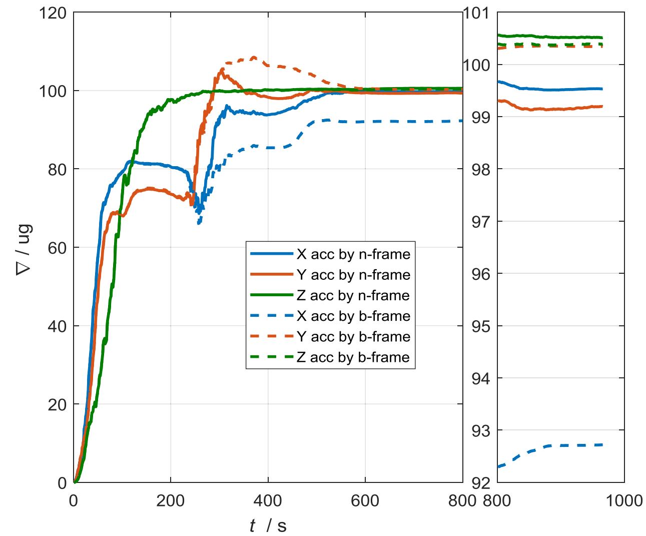

Fig. 7. Accelerometer bias estimate results.

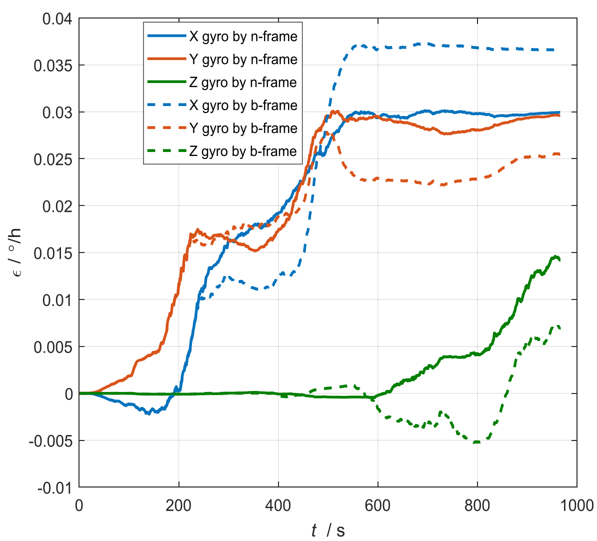

Fig. 6. Gyroscope bias estimate results.

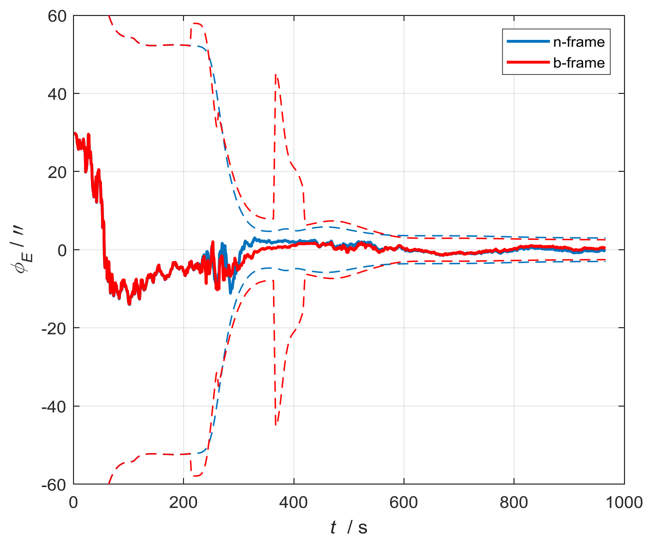

Fig. 8. Pitch estimate errors and their 3σ envelopes.

lies only in the steady-state. This means that the two types of attitude error model are both valid. The performance difference between the two attitude error models in MEKF can be explained roughly according to (27) and (50). It is shown in (27) that the coefficient of attitude error $\boldsymbol{\alpha}$ is $\left(-\hat{\boldsymbol{\omega}}_{in}^{n}\times\right)$, which is a function of the calculated velocity and position. It is shown in (50) that the coefficient of attitude error $\boldsymbol{\alpha}$ is $\left(-\hat{\boldsymbol{\omega}}_{ib}^{b}\times\right)$, which is a function of the measured body angular rate. In this respect, there is some disturbance inherent in the body frame attitude error model. Due to such disturbance, the gyroscope drift biases are estimated not so well by the MEKF with body frame attitude error, as shown in Fig.6.

In order to further evaluate the filtering performance, the two MEKF with different attitude error models are carried out one time using the same initial conditions as discussed above. The attitude errors, velocity errors, position errors and their estimated 3σ confidence intervals are plotted in Fig. 8-16. It is shown that there is some disturbance in the 3σ bounds of the MEKF with body frame attitude error model. Meanwhile, the yaw estimate error drifts outside its respective 3σ bounds by MEKF with body frame attitude error model, which indicates that corresponding filter is performing in a subpar fashion. All these results indicate that the navigation frame attitude error model is smoother than body frame attitude error model. In this respect, it is suggested to use navigation frame attitude error model in MEKF in practical applications.

B. Field Test Results With Navigation-grade INS

In order to further evaluate the MEKF for INS/GPS with different attitude error models, the field test using car-mounted equipment is carried out. A navigation-grade strapdown INS is fixed inside the car, with the GPS antenna installed outside the car on the top of the car. The field test trajectory is shown in Fig. 17. The first 2000 seconds data is used to perform INS/GPS using different algorithms. The resulting

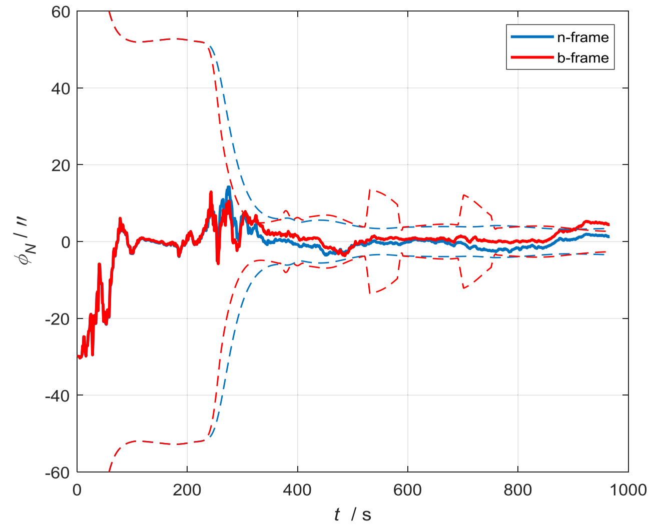

Fig. 9. Roll estimate errors and their 3σ envelopes.

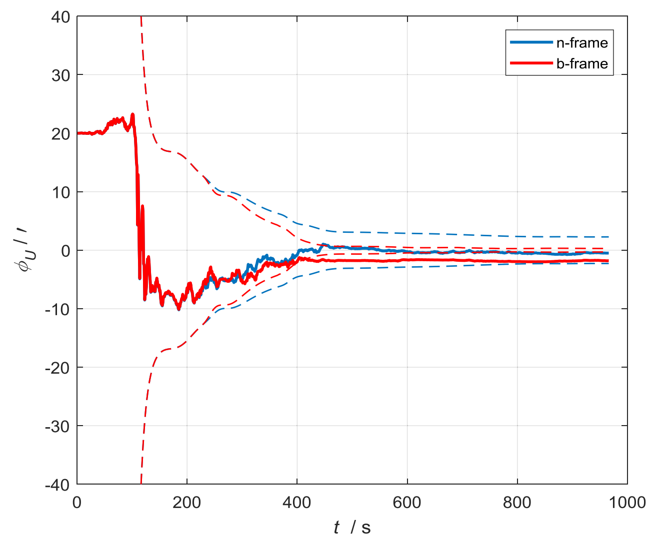

Fig. 10. Yaw estimate errors and their 3σ envelopes.

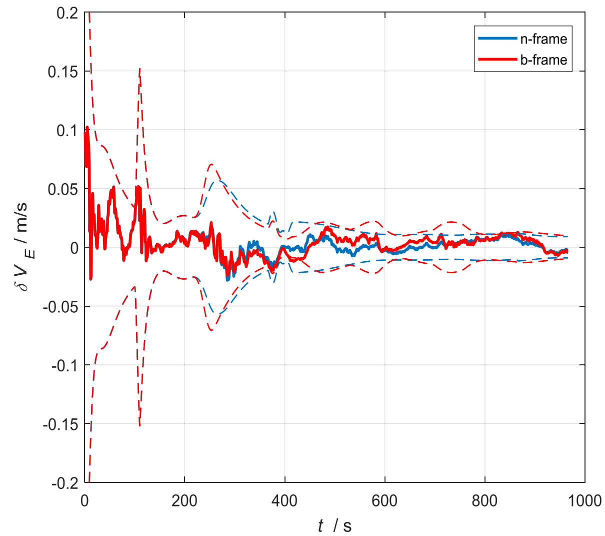

Fig. 11. East velocity estimate errors and their 3σ envelopes.

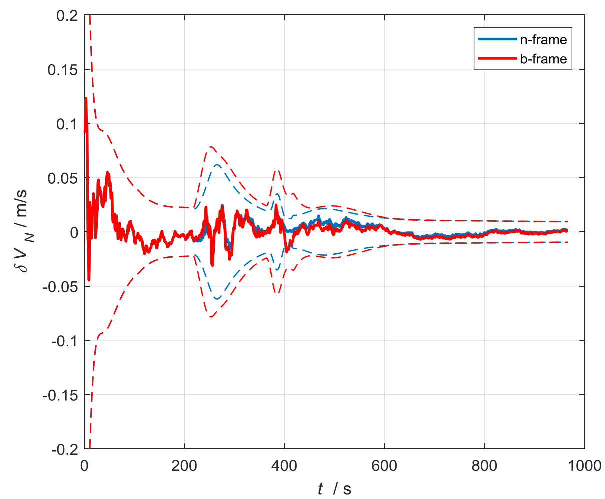

Fig. 12. North velocity estimate errors and their 3σ envelopes.

attitude estimate and inertial sensors drift biases estimate are shown in Fig. 18-23, respectively. It is shown that the attitude results by the two algorithms are very similar with each other. However, the inertial sensors drift biases estimated by MEKF with navigation frame attitude error model are smoother than those with body frame attitude error model. This is consistent with the analysis using simulated data.

Since there is no reference for the INS/GPS integration results, we cannot evaluate the estimate accuracy using the aforementioned INS/GPS integration results directly. In order to evaluate the INS/GPS integration results by the two different filters, we execute the following evaluating procedure: the estimated inertial sensors drift biases are firstly compensated for the collected gyroscopes and accelerometers measurements. Then, the optimal alignment [26]–[28] is performed using the first 300 seconds data to obtain the attitude at the very starting point. With the obtained initial attitude and velocity and position provided by GPS, pure navigation calculation is performed using 6000 seconds compensated data. The calculated position errors are used to evaluate the estimate accuracy of the inertial sensors drift biases. The consideration of such evaluation procedure is that more accurate drift biases estimate will result in more accurate calculated position. For comparison, initial alignment and pure navigation calculation without drift biases compensation are also performed. The resulting latitude error, longitude error and position error by different procedures are shown in Fig. 24-26, respectively. In these figures, "calibrated by b-frame" means that the estimated inertial sensors drift biases as shown in Fig. 21 and 23 are compensated during the initial alignment and pure navigation calculation. Similarly, "calibrated by n-frame" means that the estimated inertial sensors drift biases as shown in Fig. 20 and 22 are compensated during the initial alignment and pure navigation calculation. It is clearly shown that with inertial sensors drift biases compensation, the calculated accuracy has been much improved, which indicates that estimated inertial sensors drift biases is valid to one extent or another.

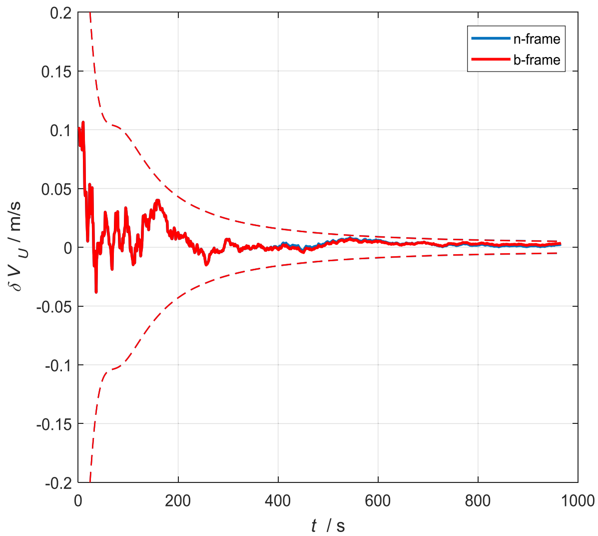

Fig. 13. Up velocity estimate errors and their 3σ envelopes.

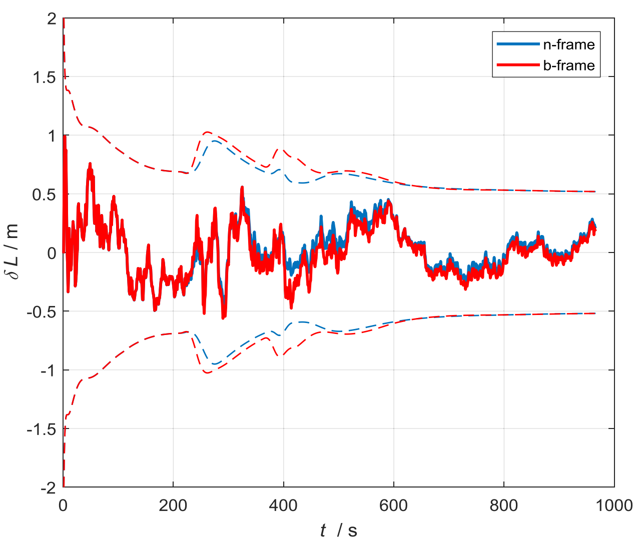

Fig. 14. Latitude estimate errors and their 3σ envelopes.

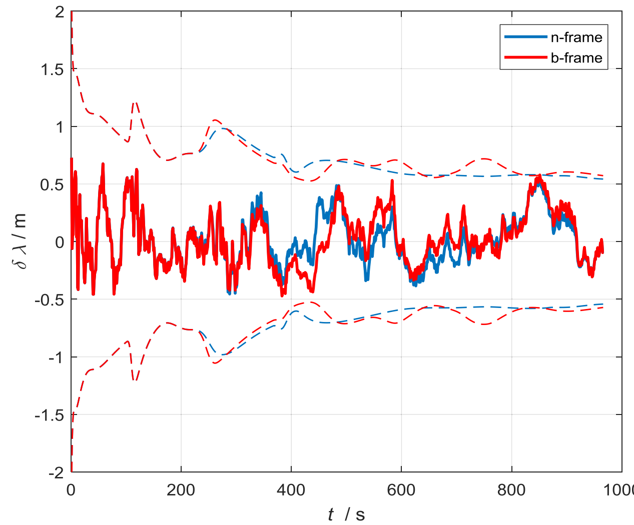

Fig. 15. Longitude estimate errors and their 3σ envelopes.

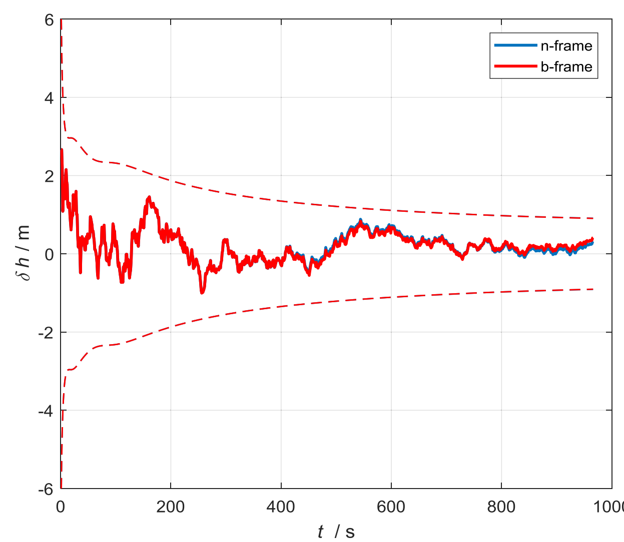

Fig. 16. Height estimate errors and their 3σ envelopes.

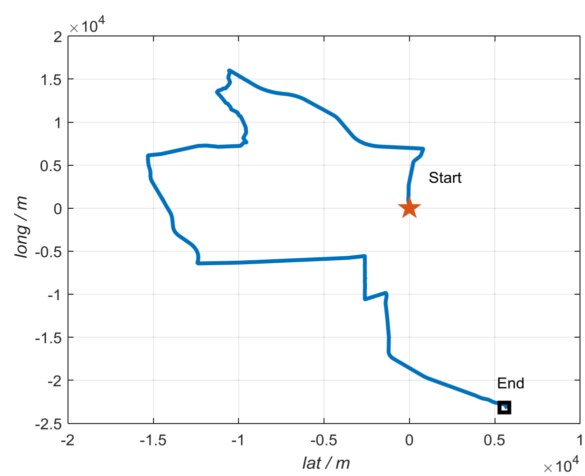

Fig. 17. Field test trajectory.

Moreover, the calculated results with inertial sensors drift biases compensation by "n-frame" is more accurate than that by "b-frame", which indicates that the estimated inertial sensors drift biases by MEKF with navigation frame attitude error model are more accurate that those with body frame attitude error model.

C. Field Test Results With Low-Cost INS

According to the aforementioned analysis and evaluation, it can be concluded that the degraded performance of the body frame attitude error model is mainly caused by the coefficient $\left(-\hat{\boldsymbol{\omega}}_{ib}^{b}\times\right)$. Intuitively, the noise or disturbance in this factor will be more severe for low-cost gyroscopes. In this respect, we would like to see the performance of the two attitude error models in low-cost INS. With this consideration, a further evaluation is carried out using low-cost strapdown INS. In this test, the experimental platform is equipped with a low-cost MEMS inertial measurements unit (IMU) and an attitude and heading reference systems (AHRS). The dominating inertial sensors errors of the low-cost IMU are given by:

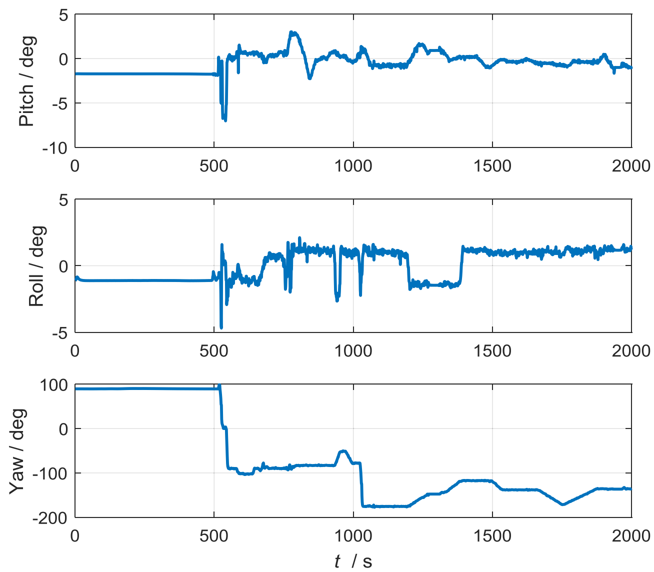

Fig. 18. Attitude estimate results with navigation frame attitude error model.

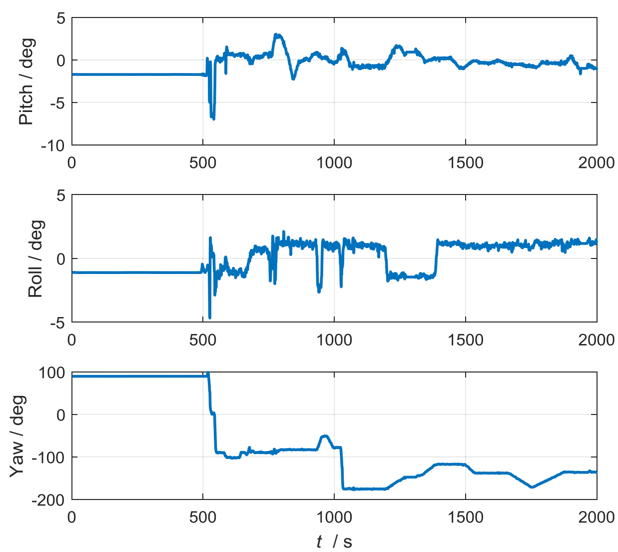

Fig. 19. Attitude estimate results with body frame attitude error.

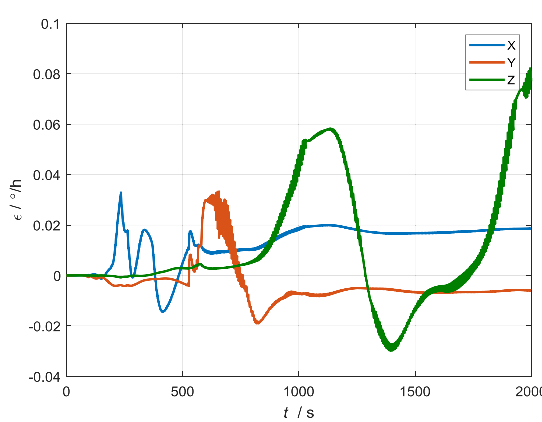

Fig. 20. Gyroscope bias estimate with navigation frame attitude error model.

gyroscopes drift bias $\leq 0.5^{\circ}/s$ and accelerometers drift bias $\leq 0.005g$. The AHRS is used to provide the reference attitude of high accuracy for comparison. Therefore, we can evaluate

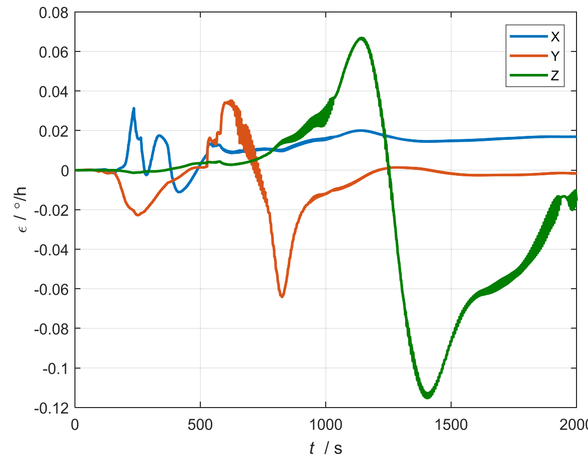

Fig. 21. Gyroscope biases estimate with body frame attitude error model.

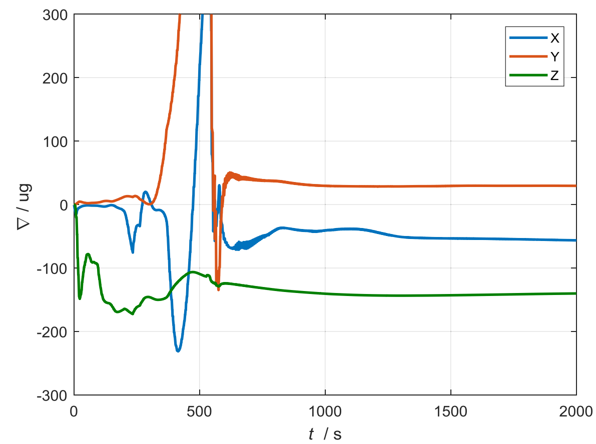

Fig. 22. Accelerometer bias estimate with navigation frame attitude error model.

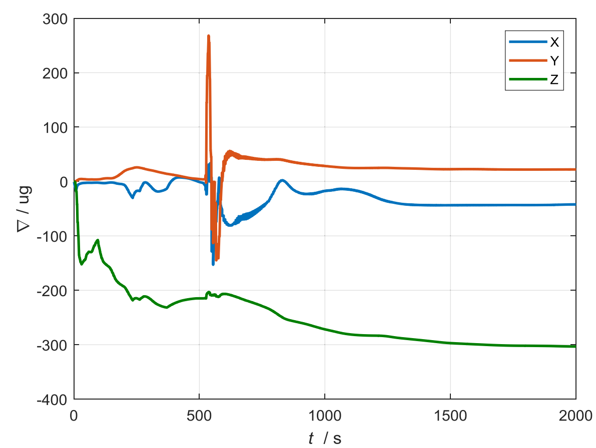

Fig. 23. Accelerometer bias estimate with body frame attitude error model.

the MEKF for INS/GPS with different attitude error models directly. The resulting attitude estimate and the corresponding estimate errors are shown in Fig. 27-29, respectively. It is clearly shown that the attitude results by the MEKF with navigation frame attitude error model are more accurate than that with body frame attitude error model. Meanwhile, there is obvious shock in the attitude results by MEKF with body

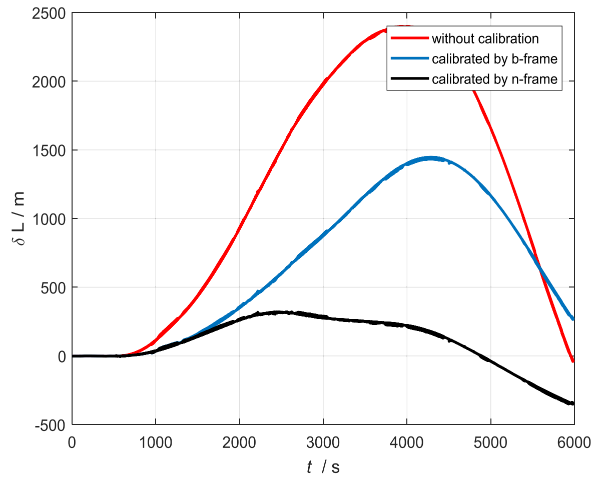

Fig. 24. Calculated latitude errors.

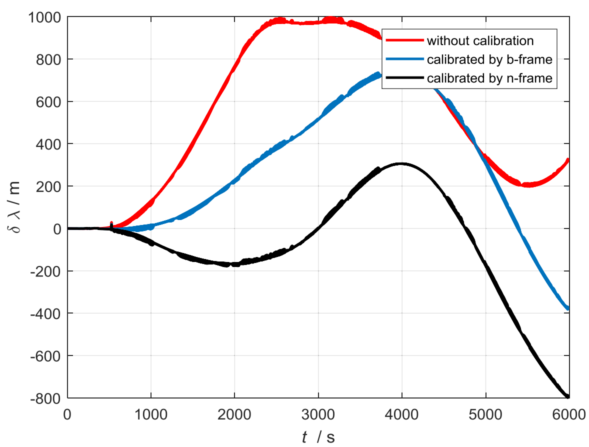

Fig. 25. Calculated longitude errors.

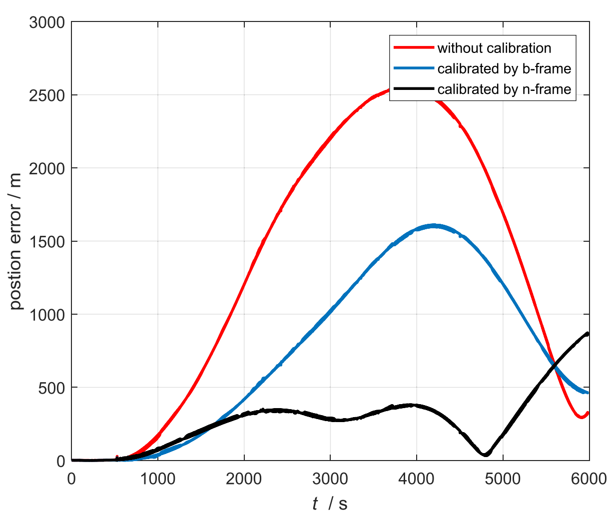

Fig. 26. Calculated position errors.

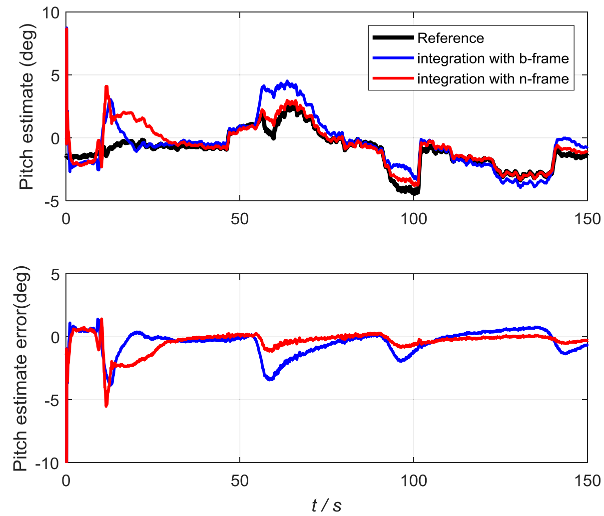

Fig. 27. Pitch estimate results with two attitude error models.

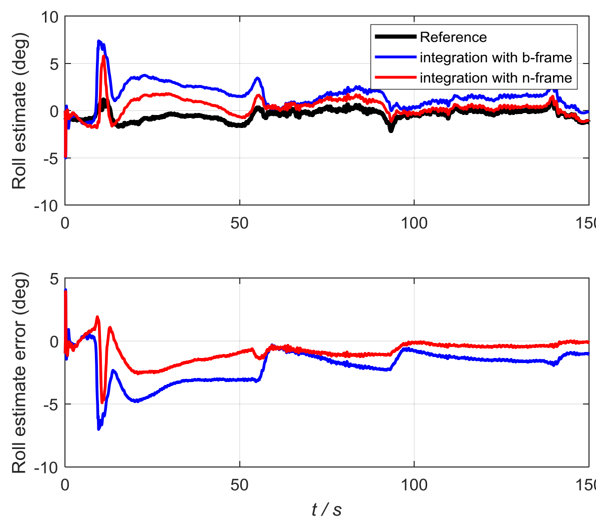

Fig. 28. Roll estimate results with two attitude error models.

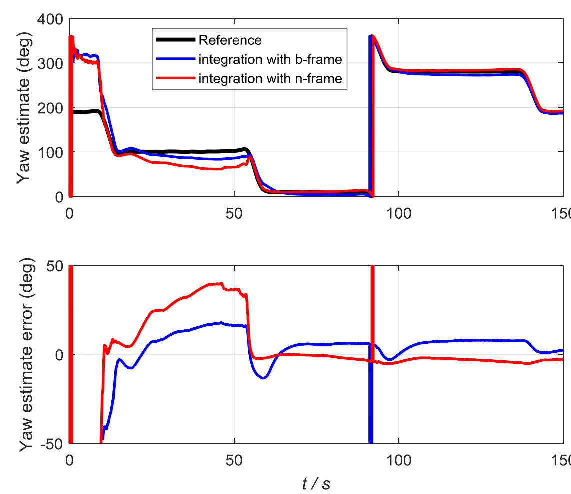

Fig. 29. Yaw estimate results with two attitude error models.

frame attitude error model as shown in Fig. 27 and 28. This is mainly due to the model disturbance caused by the coefficient $\left(-\hat{\omega}_{ib}^{b}\times\right)$ in the attitude error model.

The inertial sensors drift biases estimate are shown in Fig. 30 and 31, respectively. It is shown that the inertial sensors drift biases estimated by MEKF with navigation frame attitude

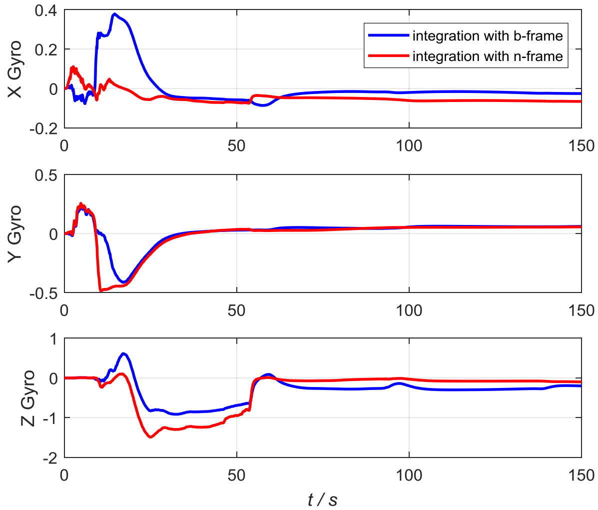

Fig. 30. Gyroscope biases estimate with two attitude error models (unit: °/*s*).

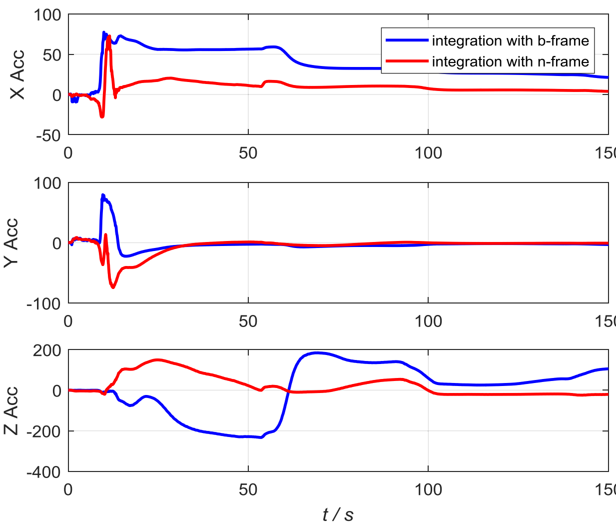

Fig. 31. Accelerometers biases estimate with two attitude error models (unit: *mg*).

error model are smoother than that with body frame attitude error model. This is consistent with the aforementioned analysis using simulated data and navigational grade INS data. Actually, the inertial sensors drift biases have a significant effect on the attitude estimate. We can also deduce that the drift biases estimate results by MEKF with navigation frame attitude error model are more accurate than those with body frame attitude error model according to Fig. 27-29.

V. Conclusion

How to parameterize the attitude error in the Kalman filtering framework is a crucial research topic in this field. Most of previous works have been devoted themselves to investigating different attitude parameterizations or introducing sophisticated new filters. In contrast, the expressed frames of the attitude errors have attracted much less attentions. In this paper, the MEKF with navigation frame attitude error is investigated for GPS/INS applications. Compared with the traditional body frame attitude error model, the navigation frame attitude error model suffers fewer disturbances. Within the MEKF framework, the navigation frame and body frame attitude error models have been evaluated thoroughly through simulations and field test data. The results show that, for navigation-grade INS the MEKF with navigation frame attitude error model performs a little better than that with body frame attitude error model, but the corresponding difference is not so obvious. This is because that in such situation, the model disturbance caused by the coefficient of the body frame attitude error is not so severe. However, for low-cost INS application, the superiority of the navigation frame attitude error model is more obvious, which is because that the aforementioned coefficient contains much more severe disturbance.

Appendix

Given the attitude error quaternion definition in (49), the derivative of $\boldsymbol{\delta}\mathbf{q}$ can be derived as

$$\boldsymbol{\delta}\dot{\mathbf{q}} = \dot{\hat{\mathbf{q}}}_b^{n^*} \otimes \mathbf{q}_b^n + \hat{\mathbf{q}}_b^{n^*} \otimes \dot{\mathbf{q}}_b^n \tag{53}$$

Substituting (16) into (53) gives

$$\begin{aligned}\boldsymbol{\delta}\dot{\mathbf{q}} &= \frac{1}{2}\hat{\mathbf{q}}_b^{n*} \otimes \mathbf{q}_b^n \otimes \boldsymbol{\omega}_{nb}^b + \frac{1}{2}\left(\hat{\mathbf{q}}_b^n \otimes \hat{\boldsymbol{\omega}}_{nb}^b\right)^* \otimes \mathbf{q}_b^n \\ &= \frac{1}{2}\hat{\mathbf{q}}_b^{n^*} \otimes \mathbf{q}_b^n \otimes \boldsymbol{\omega}_{nb}^b - \frac{1}{2}\hat{\boldsymbol{\omega}}_{nb}^b \otimes \hat{\mathbf{q}}_b^{n^*} \otimes \mathbf{q}_b^n \\ &= \frac{1}{2}\boldsymbol{\delta}\mathbf{q} \otimes \boldsymbol{\omega}_{nb}^b - \frac{1}{2}\hat{\boldsymbol{\omega}}_{nb}^b \otimes \boldsymbol{\delta}\mathbf{q}\end{aligned} \tag{54}$$

For multiplications between quaternion and three-dimensional vectors, the following equations hold

$$\mathbf{q} \otimes \boldsymbol{\omega} = \begin{bmatrix} 0 & -\boldsymbol{\omega}^T \\ \boldsymbol{\omega} & -(\boldsymbol{\omega}\times) \end{bmatrix}\mathbf{q} \tag{55a}$$

$$\boldsymbol{\omega} \otimes \mathbf{q} = \begin{bmatrix} 0 & -\boldsymbol{\omega}^T \\ \boldsymbol{\omega} & (\boldsymbol{\omega}\times) \end{bmatrix}\mathbf{q} \tag{55b}$$

Substituting (55) into (54) gives

$$\begin{aligned}\boldsymbol{\delta}\dot{\mathbf{q}} &= \frac{1}{2}\left\{\begin{bmatrix} 0 & -\boldsymbol{\omega}_{nb}^{bT} \\ \boldsymbol{\omega}_{nb}^b & -\left(\boldsymbol{\omega}_{nb}^b\times\right) \end{bmatrix} - \begin{bmatrix} 0 & -\hat{\boldsymbol{\omega}}_{nb}^{bT} \\ \hat{\boldsymbol{\omega}}_{nb}^b & \left(\hat{\boldsymbol{\omega}}_{nb}^b\times\right) \end{bmatrix}\right\}\boldsymbol{\delta}\mathbf{q} \\ &= \frac{1}{2}\begin{bmatrix} 0 & -\boldsymbol{\delta}\boldsymbol{\omega}_{nb}^{bT} \\ \boldsymbol{\delta}\boldsymbol{\omega}_{nb}^b & -\left(\boldsymbol{\delta}\boldsymbol{\omega}_{nb}^b + 2\hat{\boldsymbol{\omega}}_{nb}^b\right)\times \end{bmatrix}\boldsymbol{\delta}\mathbf{q}\end{aligned} \tag{56}$$

With the approximation in (26), (56) can be expanded as

$$\frac{\dot{\boldsymbol{\alpha}}}{2} = \frac{1}{2}\left[\boldsymbol{\delta}\boldsymbol{\omega}_{nb}^b - \frac{1}{2}\left(\boldsymbol{\delta}\boldsymbol{\omega}_{nb}^b\times\right)\boldsymbol{\alpha} - \left(\hat{\boldsymbol{\omega}}_{nb}^b\times\right)\boldsymbol{\alpha}\right] \tag{57}$$

Ignoring the high-order small value, (57) can be approximated as

$$\dot{\boldsymbol{\alpha}} = \boldsymbol{\delta}\boldsymbol{\omega}_{nb}^b - \left(\hat{\boldsymbol{\omega}}_{nb}^b\times\right)\boldsymbol{\alpha} \tag{58}$$

Substituting (17) and (22) into (58) gives

$$\begin{aligned}\dot{\boldsymbol{\alpha}} &= -\mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T \hat{\boldsymbol{\omega}}_{in}^n \times \boldsymbol{\alpha} - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T \boldsymbol{\delta}\boldsymbol{\omega}_{in}^n + \boldsymbol{\delta}\boldsymbol{\omega}_{ib}^b \\ &\quad - \left(\hat{\boldsymbol{\omega}}_{ib}^b - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T \hat{\boldsymbol{\omega}}_{in}^n\right) \times \boldsymbol{\alpha} \\ &= -\hat{\boldsymbol{\omega}}_{ib}^b \times \boldsymbol{\alpha} - \mathbf{C}\left(\hat{\mathbf{q}}_b^n\right)^T \boldsymbol{\delta}\boldsymbol{\omega}_{in}^n + \boldsymbol{\delta}\boldsymbol{\omega}_{ib}^b\end{aligned} \tag{59}$$

which is just (50).

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